UNIVERSITY OF TIRANA
FACULTY OF ECONOMY
MASTER OF SCIENCE - INFORMATION SECURITY

TIER 2: BUILDER
CREATING A STREAMLIT BI ASSISTANT

SPECIAL COURSE

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INTRODUCTION

This project presents the development of an AI-enhanced Business Intelligence OLAP Assistant designed to transform raw e-commerce data into an interactive analytical system. The dataset, scraped from Bed Bath & Beyond, initially exhibited multiple data quality issues, including missing values, inconsistent formatting, and structural irregularities, preventing reliable analysis.
The project was structured into three phases.
The first phase focused on data profiling and business requirement identification, defining product-level granularity and classifying data quality problems such as inaccuracies, inconsistencies, and incompleteness.

The second phase implemented a systematic ETL pipeline using Pentaho Spoon to clean, standardize, and enrich the dataset, producing a structured and analysis-ready file containing over 845,000 product records. This phase ensured data integrity and analytical reliability.

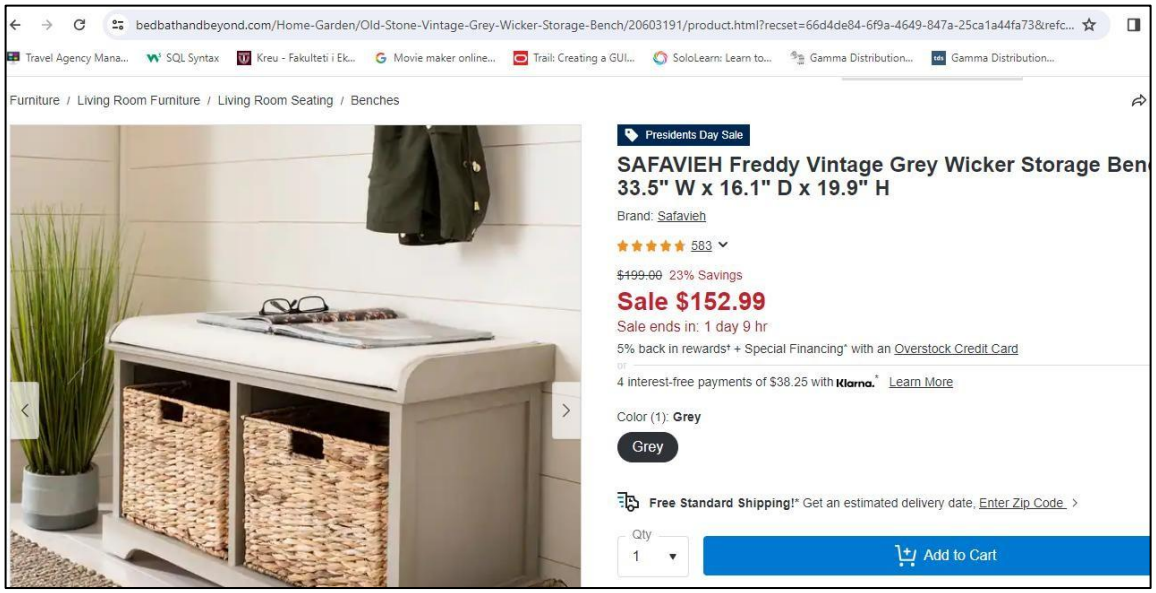
The third phase introduced an AI-powered OLAP assistant developed using Streamlit and Pandas, integrating the Claude API from Anthropic. The system translates natural language queries into executable analytical operations and supports core OLAP functionalities, including Slice, Dice, Group & Summarize, Drill-Down, and Compare. Results are dynamically displayed through interactive visualizations.

The project demonstrates the interdependence between data quality management, ETL processes, OLAP modeling, and AI-based query generation. It highlights that reliable AI-driven analytics depends fundamentally on structured data preparation and controlled prompt engineering. This work provides a practical framework for integrating generative AI within modern Business Intelligence environments.

PHASE 0

Non-Tidy Dataset - Raw data scraped from **BadBath&Beyond** website for analysing . The database has got the following fields:

1. **attributes -** *All information of the product including item number, specifications, features parsed by JavaScript;*
2. **availability -** *Whether the product is in stock of out or stock at the moment of the scrape;*
3. **brand -** *The brand of the product;*
4. **category -** *The breadcrumb of how navigated to the website for the specific product;*
5. **description -** *The description shown on the website;*
6. **language -** *Whether it is **en** (English) or **fr** (French);*
7. **model -** *The model number of the product;*
8. **name -** *Product headline shown on website;*
9. **productdetails -** *All product information parsed by JavaScript;*
10. **productid -** *The unique product number;*
11. **regular price -** *List price or regular price of the product, never empty;*
12. **sale price -** *The price applied on sale of the product.*



This dataset contains mostly qualitative data, which need to be cleaned for preparing a clean and tidy database to be sent to the client to analyse those data for its own purposes.

1) **Market Research:** Clients can analyze product listings, pricing information, and customer reviews to gain insights into market trends, competitor strategies, and consumer preferences. This can help them make informed decisions about product development, pricing strategies, and marketing campaigns.

2) **Competitor Analysis:**. This information can be used to identify gaps in the market, benchmark against competitors, and refine business strategies.

3) **Price Monitoring and Optimization:** Clients can track price changes for specific products over time and across competitors. This data can be leveraged to optimize pricing strategies, identify opportunities for discounts or promotions, and ensure competitiveness in the market.

4) **Inventory Management:** Can provide insights into product availability, stock levels, and inventory turnover rates. Clients can use this information to optimize their inventory management processes, avoid stockouts or overstocking, and improve supply chain efficiency.

5) **Product Recommendations and Personalization:** Analysing customer reviews and purchase histories can help clients understand consumer preferences and behaviour. This data can be used to enhance product recommendations, personalize marketing messages, and improve the overall customer experience.

6) **Content Creation and SEO:** Clients can extract product descriptions, specifications, and images from websites to create content for their own e-commerce platforms, blogs, or marketing materials. Additionally, keyword analysis can inform search engine optimization (SEO) strategies to improve online visibility and drive organic traffic.

7) **Ad Targeting and Retargeting:** Insights gained can inform targeted advertising campaigns on platforms like Google Ads or social media. By understanding consumer preferences and behaviours, clients can create more relevant ad content, improve ad targeting accuracy, and increase conversion rates.

8) **Product Development and Innovation:** Analysing product features, customer feedback, and emerging trends can inspire product innovation and development. Clients can identify opportunities to introduce new products or improve existing ones based on market demand and customer preferences.

   But in order to do this we need **CLEAN DATA.**

PHASE 1
DATA PROFILING AND CLEANING FOR ENHANCED DATA QUALITY

1) Identify business process requirements

a. Document and prioritize key business processes to be analysed.

- **Inventory Management:** Understanding product availability (in stock or out of stock) is crucial for inventory management. It helps in determining stock levels, replenishment needs, and forecasting demand.
- **Pricing Strategy:** Analysing regular and sale prices can provide insights into pricing strategies. It helps in assessing the effectiveness of discounts, promotions, and overall pricing competitiveness.
- **Product Categorization:** Analysing product categories and how customers navigate through the website (breadcrumb) can help in optimizing website navigation, improving user experience, and organizing products effectively.
- **Language Localization:** Understanding the distribution of products across different languages (English and French) can help in tailoring marketing strategies and content localization efforts.
- **Product Description Analysis:** Analysing product descriptions can provide insights into product features, specifications, and customer preferences. It helps in optimizing product descriptions for better visibility and conversion.
- **Brand Analysis:** Understanding the distribution of products across different brands can help in assessing brand performance, identifying popular brands, and optimizing brand partnerships.
- **Sales Performance Analysis:** Analysing regular and sale prices alongside product availability can provide insights into sales performance, revenue generation, and overall business profitability.
- **Data Quality Assurance:** Ensuring data accuracy, completeness, and consistency is crucial for reliable analysis and decision-making. This involves identifying and addressing data quality issues such as missing values, duplicates, and inconsistencies.

b. Functional Requirements - *Specify the functionalities expected from the data in alignment with business processes.*

- **Inventory Management**
  - **Functionality:** Provide accurate and up-to-date information on product availability (in stock or out of stock).
  - **Requirement:** Ensure that the 'availability' field accurately reflects the current stock status of each product.
  - **Validation:** Regularly update availability status based on inventory changes and ensure synchronization with the actual stock levels.

- **Pricing Strategy**
  - **Functionality:** Analyze regular and sale prices to assess pricing strategies and competitiveness.
  - **Requirement:** Ensure that both 'regular price' and 'sale price' fields are populated accurately for each product.
  - **Validation:** Verify consistency between regular and sale prices, identify discrepancies or anomalies, and validate pricing changes against promotional activities.

- **Product Categorization**
  - **Functionality:** Analyze product categories and navigation patterns to optimize website organization and user experience.
  - **Requirement:** Ensure that the 'category' field provides clear and consistent categorization for each product.
  - **Validation:** Validate the accuracy of product categorization by comparing it with the website's navigation structure and identifying any inconsistencies or missing categories.

- **Language Localization**
  - **Functionality:** Tailor marketing strategies and content localization efforts based on language preferences.
  - **Requirement:** Ensure that the 'language' field accurately indicates the language of each product (English or French).
  - **Validation:** Validate language localization by cross-referencing product descriptions and other text fields with the specified language indicator.

- **Product Description Analysis**
  - **Functionality:** Analyze product descriptions to understand product features, specifications, and customer preferences.
  - **Requirement:** Ensure that the 'description' field provides comprehensive and informative descriptions for each product.
  - **Validation:** Verify the completeness and accuracy of product descriptions, identify any missing or misleading information, and ensure consistency across products.

- **Brand Analysis**
  - **Functionality:** Assess brand performance and popularity among customers.
  - **Requirement:** Ensure that the 'brand' field accurately identifies the brand associated with each product.
  - **Validation:** Validate brand attribution by cross-referencing with official brand information, identifying any inconsistencies or misspellings.

- **Sales Performance Analysis**
  - **Functionality:** Analyze sales performance based on pricing and availability data.
  - **Requirement:** Ensure that pricing and availability data are accurately captured and synchronized.
  - **Validation:** Monitor sales trends, identify any discrepancies or anomalies in pricing or availability, and investigate factors influencing sales performance.

- **Data Quality Assurance**
  - **Functionality:** Ensure data accuracy, completeness, and consistency for reliable analysis and decision-making.
  - **Requirement:** Implement data cleaning and validation procedures to address issues such as missing values, duplicates, and inconsistencies.
  - **Validation:** Conduct regular data quality checks, validate data against predefined quality metrics, and implement corrective actions as needed to maintain data integrity.

c. **Non-Functional Requirements -** *Identify criteria such as data accuracy, completeness, and relevance to meet business objectives*

- **Data Accuracy**
  - **Requirement**: The data must accurately represent the real-world attributes, availability, pricing, and other information related to the products listed on the **BadBath&Beyond** website.
  - Criteria:
    - Accuracy of product availability status (in stock or out of stock).
    - Precision in capturing regular and sale prices without errors or discrepancies.
    - Correctness of product descriptions, specifications, and other details.
    - Accurate attribution of brands and categorization of products.

- **Data Completeness**
  - **Requirement:** The data must contain all necessary fields and information required for analysis and decision-making.
  - Criteria:
    - All products should have complete information on attributes, availability, pricing, brand, category, description, language, model, and product details.
    - No missing values or incomplete records in any of the essential fields.
    - Complete coverage of products across different categories and brands.

- **Data Relevance**
  - Requirement: The data must be relevant to the business objectives and decision-making processes.
  - Criteria:
    - Relevance of product attributes, descriptions, and other details to customer needs and preferences.
    - Alignment of product categorization with customer navigation patterns and website organization.
    - Language localization should match the target audience's preferences and support effective marketing and communication strategies.
    - Brands and products included in the dataset should be relevant to the business's target market and customer base.

- **Data Consistency**
  - **Requirement:** The data must maintain consistency across different fields and records.
  - Criteria:
    - Consistency in formatting and structure across all data fields.
    - Consistent representation of product attributes, pricing, and availability status.
    - Uniformity in language localization indicators (e.g., consistent use of language codes).
    - Consistent attribution of brands and categorization of products across records.

- **Data Timeliness**
    - **Requirement:** The data must be updated and refreshed regularly to reflect current information.
    - Criteria:
        - Timeliness of updates to product availability status to reflect changes in stock levels.
        - Regular updates to pricing information to capture changes in regular and sale prices.
        - Timely incorporation of new products and removal of discontinued items from the dataset.

2) Identify the problem

a) Define the level of granularity at which we want to analyze the data, ensuring it aligns with business objectives.

Based on the considerations provided and the nature of the data from the **BadBath&Beyond** website, Analysing the data at the product level would be the most appropriate granularity level. This choice aligns well with the identified business objectives and facilitates meaningful insights across various areas such as inventory management, pricing strategy optimization, sales performance analysis, and data quality assurance.

Analysing data at the product level allows for a detailed examination of individual products' attributes, availability, pricing, and performance. This granularity level enables stakeholders to gain actionable insights into specific products, identify trends, make informed decisions, and optimize business strategies accordingly.

Furthermore, analysing at the product level ensures that the data is sufficiently detailed to meet decision-making needs while remaining practical and efficient for analysis. It also allows for effective data quality assurance measures to maintain the accuracy, completeness, consistency, and relevance of the data.

Therefore, analysing data at the product level would best serve the business objectives and facilitate informed decision-making processes for **BadBath&Beyond**.

b) Consider the specific challenges posed by textual data, such as unstructured content, linguistic nuances, and context.

- **Unstructured Content:** Textual data often lacks a predefined structure, making it challenging to extract meaningful insights directly. Product descriptions may vary widely in length, style, and formatting, making it difficult to standardize analysis.

- **Linguistic Nuances:** Natural language is inherently nuanced, with variations in grammar, syntax, vocabulary, and semantics. Different linguistic nuances, such as idiomatic expressions, slang, or regional dialects, can affect the interpretation of text data and require specialized handling.

- **Contextual Understanding:** Textual data often requires context to be properly interpreted and analyzed. Understanding the context in which the text was written, including the intended audience, purpose, and cultural background, is essential for accurate analysis.

- **Ambiguity and Polysemy:** Textual data may contain ambiguous terms or words with multiple meanings (polysemy), leading to potential misinterpretation. Resolving ambiguity and disambiguating polysemous words requires context-aware analysis and advanced natural language processing techniques.

- **Noise and Irrelevant Information:** Textual data often contains noise or irrelevant information that can obscure meaningful insights. This includes extraneous characters, punctuation, formatting artifacts, as well as irrelevant or redundant text.

- **Data Preprocessing Challenges:** Preprocessing textual data involves various tasks such as tokenization, stemming, lemmatization, stop-word removal, and part-of-speech tagging.

- **Sentiment and Opinion Analysis:** Analysing sentiment and opinions expressed in textual data adds another layer of complexity. Understanding the sentiment polarity (positive, negative, neutral) and discerning subjective opinions from factual information requires specialized techniques such as sentiment analysis and opinion mining.

- **Data Sparsity:** Textual data often exhibits data sparsity, where many words or phrases occur infrequently or are unique to specific documents. This can affect the reliability of statistical models and require techniques such as feature engineering or dimensionality reduction to address.

- **Domain-Specific Terminology:** Textual data in certain domains may contain specialized terminology or jargon that is not commonly found in general language corpora. Understanding and processing domain-specific language requires specialized knowledge or domain-specific dictionaries.

- **Language Translation:** When dealing with multilingual textual data, challenges arise in accurately translating text between languages while preserving meaning and context. Machine translation systems may struggle with idiomatic expressions, cultural nuances, and language-specific syntax.

- **Document Length Variation:** Textual documents can vary widely in length, from short product titles to lengthy descriptions or reviews. Analysing documents of different lengths requires techniques for handling variable-length inputs, such as padding, truncation, or hierarchical modelling.

- **Temporal Dynamics:** Textual data collected over time may exhibit temporal dynamics, where language use, topics, or sentiment change over different time periods. Analysing temporal trends and detecting shifts in language patterns requires time-series analysis techniques and consideration of temporal context.

3. Profiling data

a. Provide comprehensive profiling of the dataset.

b. Understand patterns, structures and anomalies within the text data to inform subsequent cleaning and transformation steps.

1. **attributes:** Contains various product attributes such as item number, specifications, and features parsed by JavaScript.
2. **availability:** Indicates whether the product is in stock or out of stock.
3. **brand:** Represents the brand of the product.
4. **category**: Provides the breadcrumb of how the user navigated to the website for the specific product.
5. **description:** Offers the description shown on the website for each product.
6. **language:** Specifies whether the language is English (en) or French (fr).
7. **model:** Represents the model number of the product.
8. **name:** Provides the product headline shown on the website.
9. **productdetails:** Contains all product information parsed by JavaScript.
10. **productid:** Represents the unique product number.
11. **regular price:** Indicates the list price or regular price of the product.
12. **sale price:** Represents the price applied on sale of the product.

Data Types:

attributes: string

availability: string

category: string

language: string

productdetails: string

regular price: Numeric (currency format).

model: string

brand: string

description: string

name: string

productid: numeric

sale price: Numeric (currency format)

Data Quality Issues:

1) **Missing Values:** Check for missing values in all fields and handle accordingly.
2) **Outliers:** Identify outliers in numeric fields such as regular price and sale price.
3) **Inconsistent Data:** Checking for inconsistencies in fields like availability (e.g., different spellings or variations of "In Stock").
4) **Language Consistency:** Ensure consistency in language representation (e.g., "en" or "English").
5) **Duplicate Records:** Identify and remove any duplicate records based on unique identifiers like productid.

Data Distributions:

1) **Availability:** Check the distribution of products between in stock and out of stock.
2) **Price Distribution:** Examine the distribution of regular prices and sale prices to identify pricing patterns.
3) **Brand Distribution:** Analyze the distribution of products across different brands.
4) **Language Distribution:** Examine the distribution of products between English and French languages.

Text Analysis:

1) **Description Length:** Analyze the distribution of description lengths to understand the variability.
2) **Keyword Analysis:** Perform keyword extraction or frequency analysis on product descriptions to identify common themes or keywords.

Relationships:

1) **Price vs. Availability:** Exploring the relationship between price (regular and sale) and availability.
2) **Brand vs. Price:** Investigating if there's a correlation between brand and pricing.

4. Identify the data quality problems

a. Systematically identify and document issues affecting the quality of our data.

1. Missing Values

Checking for missing values in all fields, particularly in essential fields such as product name, availability, regular price, and sale price.

2. Inaccuracies

Reviewing numerical fields (regular price, sale price) for outliers or unrealistic values that do not align with expectations.

Examining textual fields (product name, description) for inaccuracies such as misspellings, grammatical errors, or misleading information.

Verifying the consistency between product attributes (model numbers, specifications) parsed from JavaScript and those provided in other fields.

3. Inconsistencies

Checking for inconsistencies in categorical fields such as availability (e.g., variations of "In Stock" or "Out of Stock") and language representation (e.g., "en" or "English").

Ensuring consistency in data formatting across fields (e.g., date formats, currency formats).

4. Incompleteness

Identifying fields with high rates of missing values or incomplete information, such as product details or specifications.

Checking for incomplete product descriptions or missing essential information that may impact the understanding of the product.

b. Classify problems such as innacuracies, inconsistencies and incompleteness.

1. Inaccuracies

Examples: *Unrealistic prices, misspelled product names, inaccurate product descriptions, incorrect availability status.*

Impact: *Misleading information, inaccurate analysis, and decision-making.*

2. Inconsistencies

Examples: *Inconsistent availability labels, variations in language representation, inconsistent formatting of data.*

Impact: *Difficulty in data integration, challenges in analysis and interpretation.*

3. Incompleteness

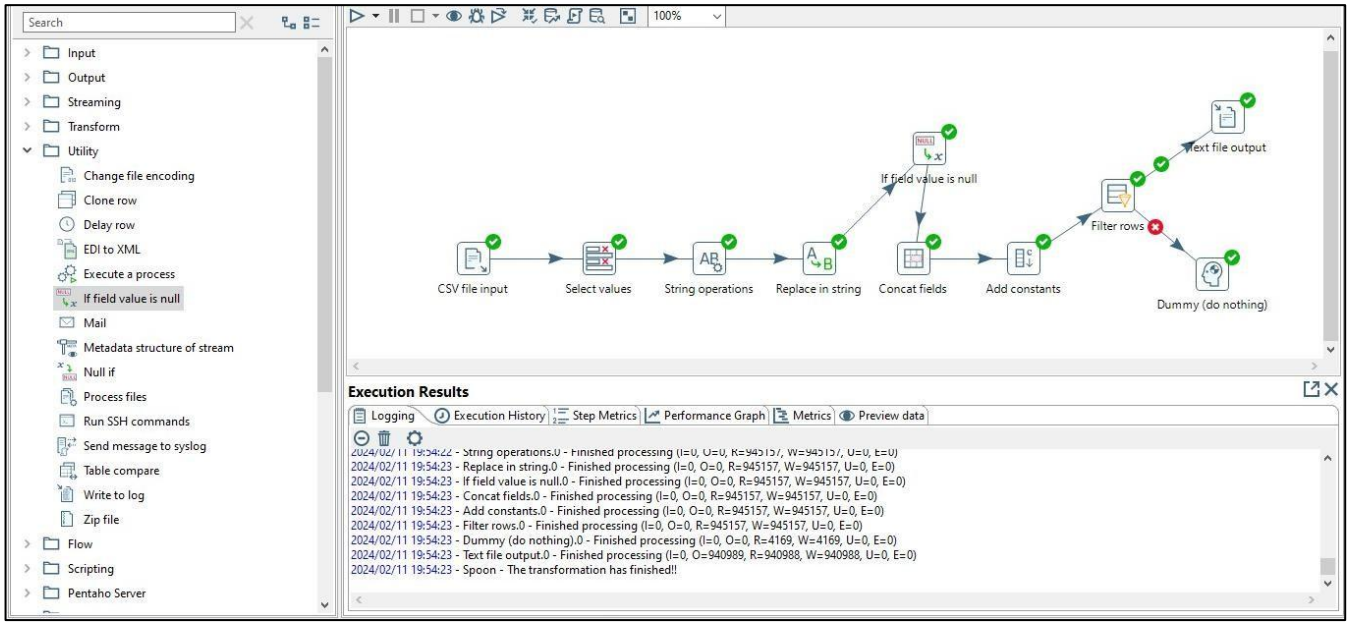
Examples: *Missing values in essential fields, incomplete product descriptions, lack of product details.*

Impact: *Reduced reliability of analysis, incomplete understanding of products, and potential bias in results.*

PHASE 2

TRANSFORMATION/ DATA CLEANING

Transformation Process:



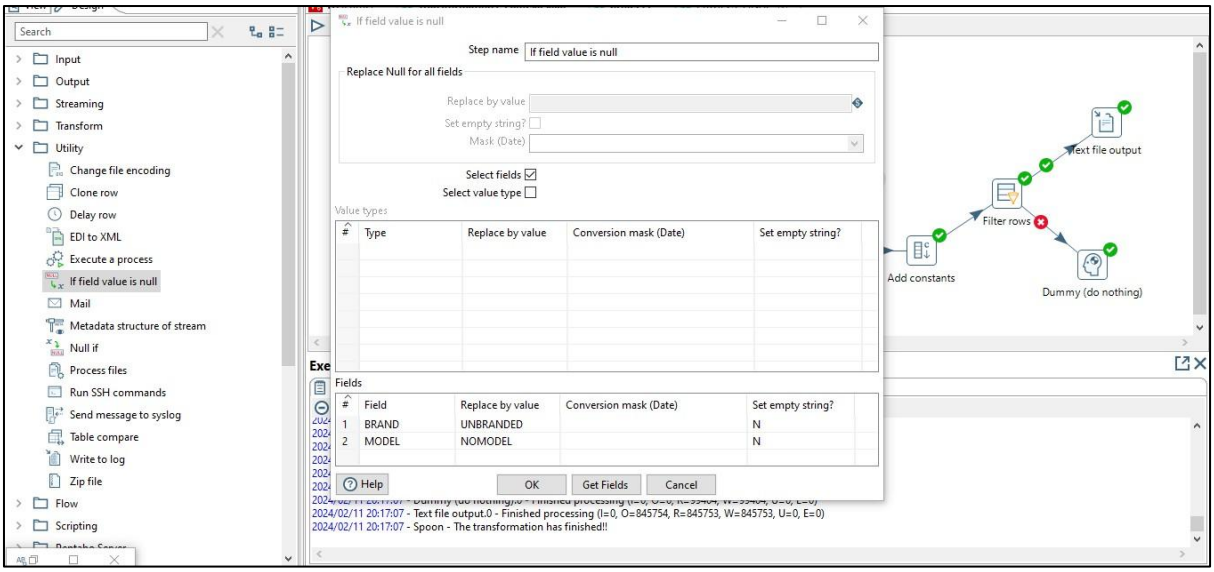
After inputting the non - tidy database populated with raw and non - tidy data gathered from web scraping, it is now ready to be cleaned and generated a clean file ready for analysis and other steps.

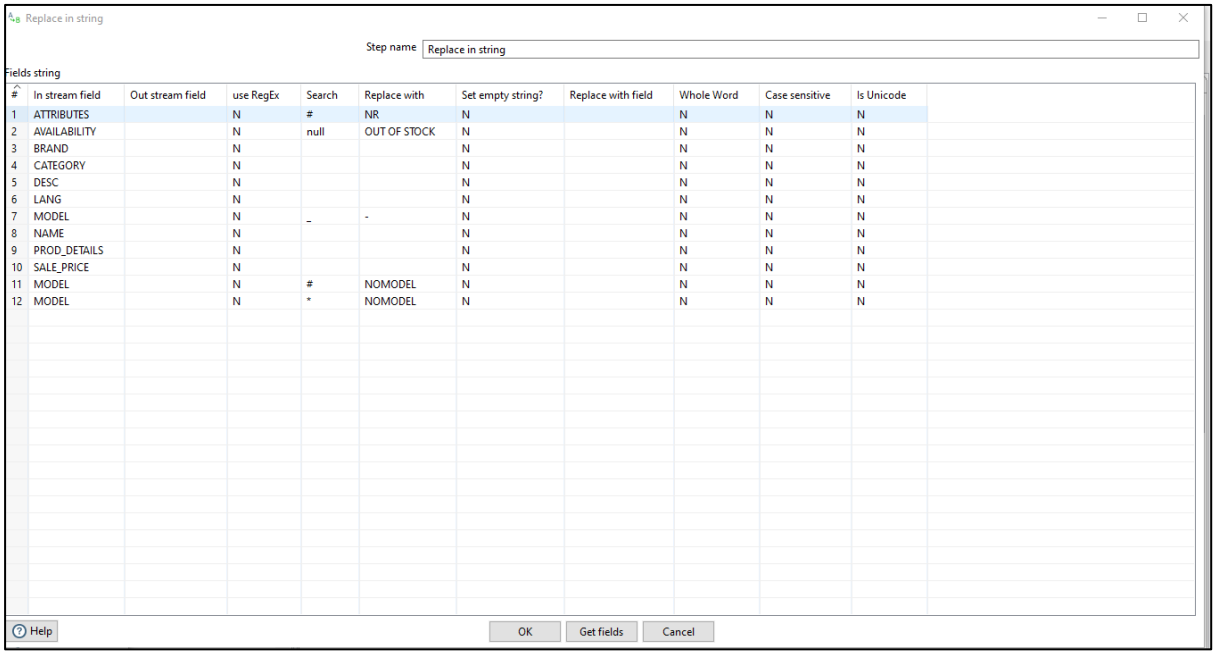
Step 1: Select values - Here I have renamed some of the fields for better reading and simplifying the process of the client to query over the data.

Step 2: String operations - I have trimmed fields from both sides and Uppercased the Availability, Brand, Lang and Model fields as required from the client to meet his criteria of processing the file.

Step 3: Replace in String - Replaced underscore with hyphen as underscore sometimes cannot be processed properly,and where there appears strange characters in Model field, replace them with NOMODEL value.

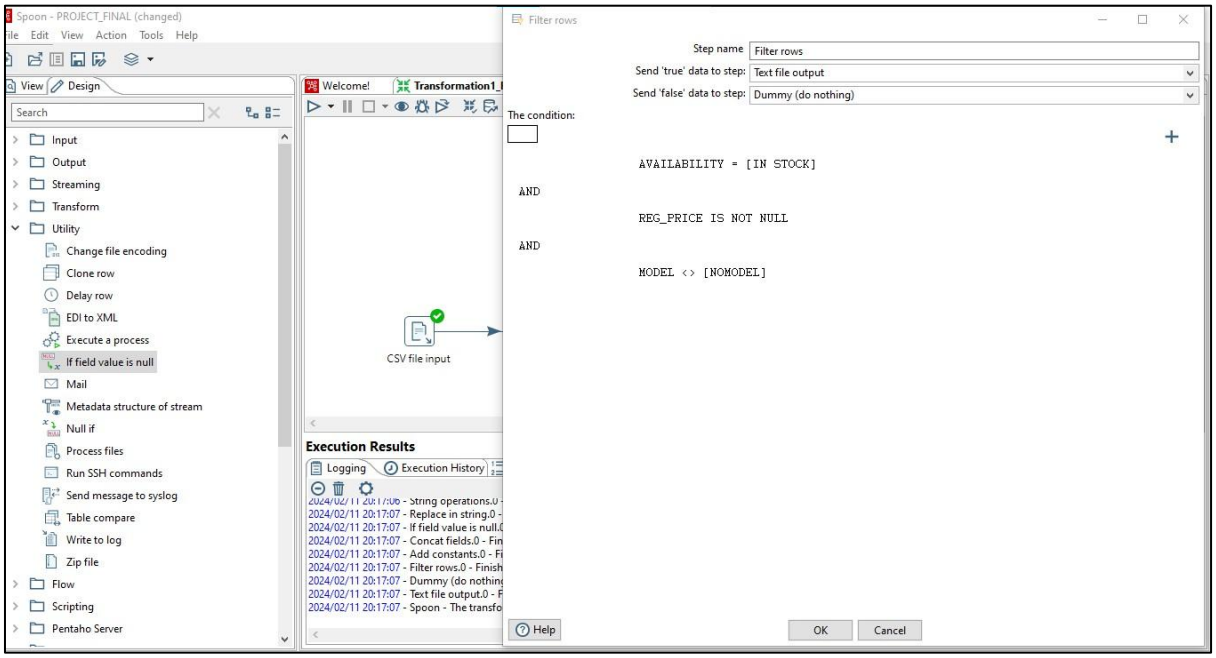
Step 4: If Field value is NULL -There happens to have null values in fields like Model & Brand and we do not want to clear up this rows either keep them empty, there happens for different products on the website to not be specified a value for the brand and model and we just replace them with NOMODEL and UNBRANDED, so the client can analyse and see which are these products.





Step 5: Concat Fields - Here I have Concatenated the fields of two values the name and productid, and created VARIANT NAME. There may be products with the same name and after duplicating we may loose this products, but if we have the variant name we will not loose products, as it is a concatenation of name+ product id which is unique for each product.

Step 6: Add constants - Here I have added the constant of Website, the scrape date and the client who is going to be delivered the data to. These constants help on analysis and differencing data of a website, because we may be cleaning datasets of different websites scraped and we need to identify which those data refer to.



Step 7: Filter Rows - We are asked to keep only the products actually in stock and we don't need the out of stock ones. So, we filter the data of where AVAILABILITY= IN STOCK and where REG_PRICE is not null. We need the products with prices to get an analysis of the prices and offers, and where MODEL is different from NOMODEL, because non - modeled products are out of stock on the site.

Step 8: Text file output - This step concludes all the cleaning process of our data. After filtering the data the way specified above we send these data to a csv file to be ready to be sent to the client for analysis.

PHASE 3

OLAP Business Intelligence Assistant

The OLAP Assistant allows business users to perform five core analytical operations through a web browser using plain English questions. The AI interprets each question, generates the correct Pandas analysis code, executes it on the real dataset, and displays the results as formatted tables and interactive charts.

User Question Example	OLAP Operation	What the App Does
How many products are in each category?	GROUP & SUMMARIZE	Groups by CATEGORY, counts products, shows bar chart
Show only products that are on sale	SLICE	Filters to rows where SALE_PRICE is not null
Show out of stock Bath products	DICE	Filters on AVAILABILITY and CATEGORY simultaneously
Break down largest category by subcategory	DRILL-DOWN	Parses breadcrumb, shows category hierarchy
Compare top 10 brands by average price	COMPARE	Groups by BRAND, calculates avg REG_PRICE, ranks top 10

1. **Technical Stack**

- **Frontend** - Streamlit
- **Data Processing** - Pandas
- **AI Integration** - Anthropic Claude API (claude-sonnet-4-20250514)
- **Visualization** - Plotly
- **Data Source** - Complete.csv (845,753 products) The cleaned **BedBath&Beyond** product catalog from Phase 2

2. **System Architecture**

The application follows a clear pipeline from user question to displayed result:

1. User types a natural language question in the browser chat box
2. app.py sends the question to the Anthropic Claude API
3. The system prompt in prompts.py tells Claude about the dataset columns and OLAP operations
4. Claude returns a JSON object containing: operation type, explanation, pandas code, chart type
5. data_utils.py executes the pandas code safely on the real dataframe
6. The result DataFrame is displayed as an interactive table
7. A Plotly chart is generated automatically if chart_type is bar, pie, or line

3. File Structure

app.py- Main Streamlit application — UI, API calls, result display

prompts.py - System prompt with dataset description and OLAP operation definitions

data_utils.py - Data loading (cached), code execution, dataset summary statistics

requirements.txt - Python package dependencies

data/Complete.csv - Cleaned BedBath&Beyond dataset (845,753 products)

.streamlit/secrets.toml - Anthropic API key storage (never submitted)

docs/README.md - Setup and usage instructions

docs/prompt_design.md - Prompt engineering strategy and iterations

4. Dataset Statistics (After Cleaning)

Metric	Value
Total Products	845,753
Unique Brands	2,680
Unique Categories	1,144
Products On Sale	520,625 (61.6%)
Average Regular Price	$332.22
Languages	English (en) and French (fr)

5. Prompt Engineering Strategy

The system prompt in prompts.py is the core of the application. It was designed to do three things:

1. Tell Claude exactly which columns exist in the dataset and their data types.
2. Define each OLAP operation with a concrete Pandas code example.
3. Force Claude to always return a structured JSON object for reliable parsing.

Every response from Claude follows this exact structure:

Required Claude Response Format

```
{
  "operation": "slice | dice | group | drilldown | compare",
  "explanation": "One sentence describing the analysis",
  "code": "pandas code that produces a variable called result",
  "chart_type": "bar | pie | line | none" }
```

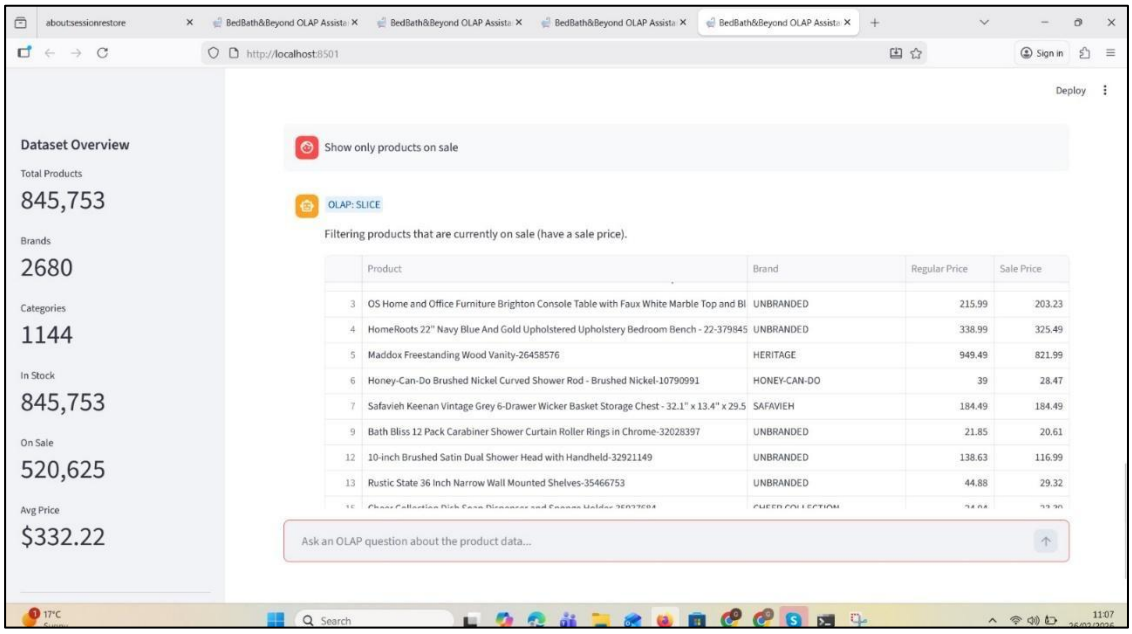
6. Problems found and fixed

- Dataset columns are uppercase (REG_PRICE) but prompt used lowercase:
  - Updated all column references to match actual dataset: BRAND, CATEGORY, REG_PRICE, SALE_PRICE, AVAILABILITY, LANG
- Claude sometimes returned JSON wrapped in markdown backticks
  - Added explicit rule: Return ONLY the JSON object. No intro text. No markdown. No backticks.
- Generated code did not always create a variable named result
  - Added prominent rule: ALWAYS assign the final output to a variable named result
- Category breadcrumb parsing was inconsistent
  - Added explicit example: df['level1'] = df['CATEGORY'].str.split('>').str[0].str.strip()
- Too many rows returned making results unreadable
  - Added rule: Always limit results to top 15 rows using .head(15)

7. OLAP Operations Examples

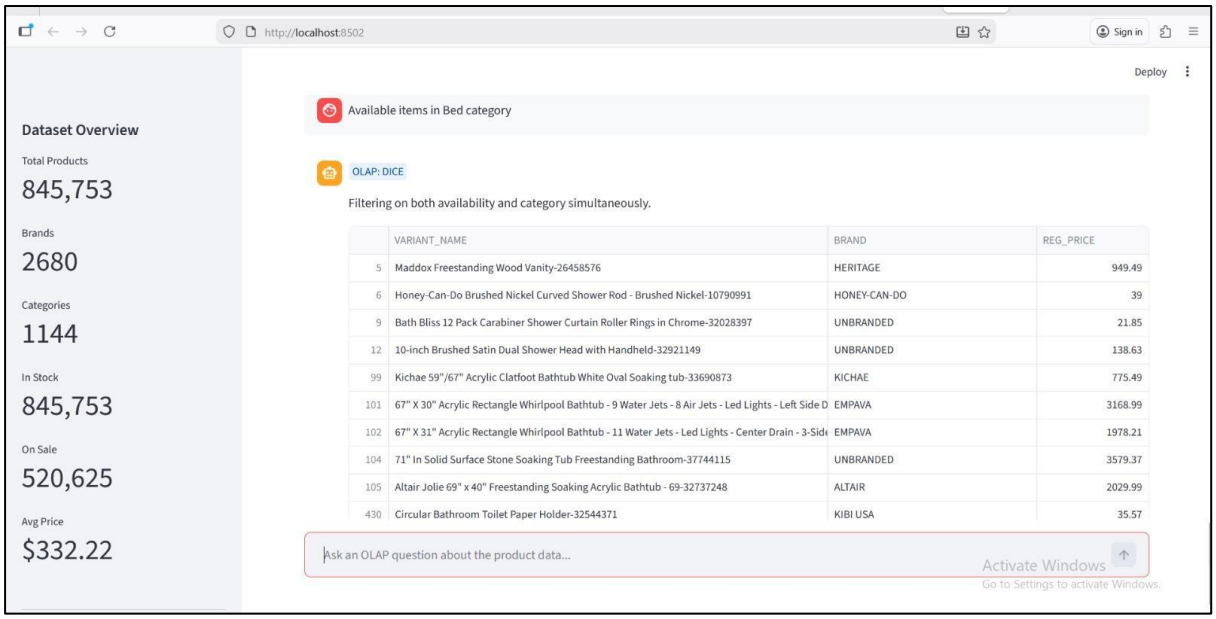
Operation 1: SLICE

Filters the dataset on a single dimension. Example: showing only products currently on sale.



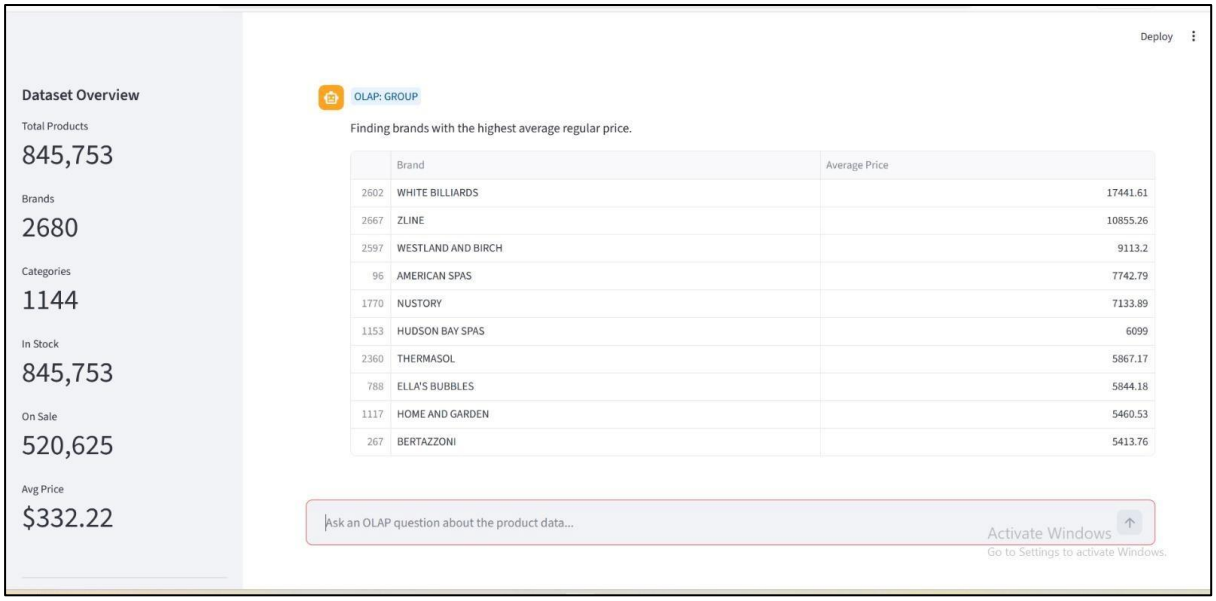
Operation 2: DICE

Filters on multiple dimensions simultaneously. Example: products in a specific category with a specific availability status.



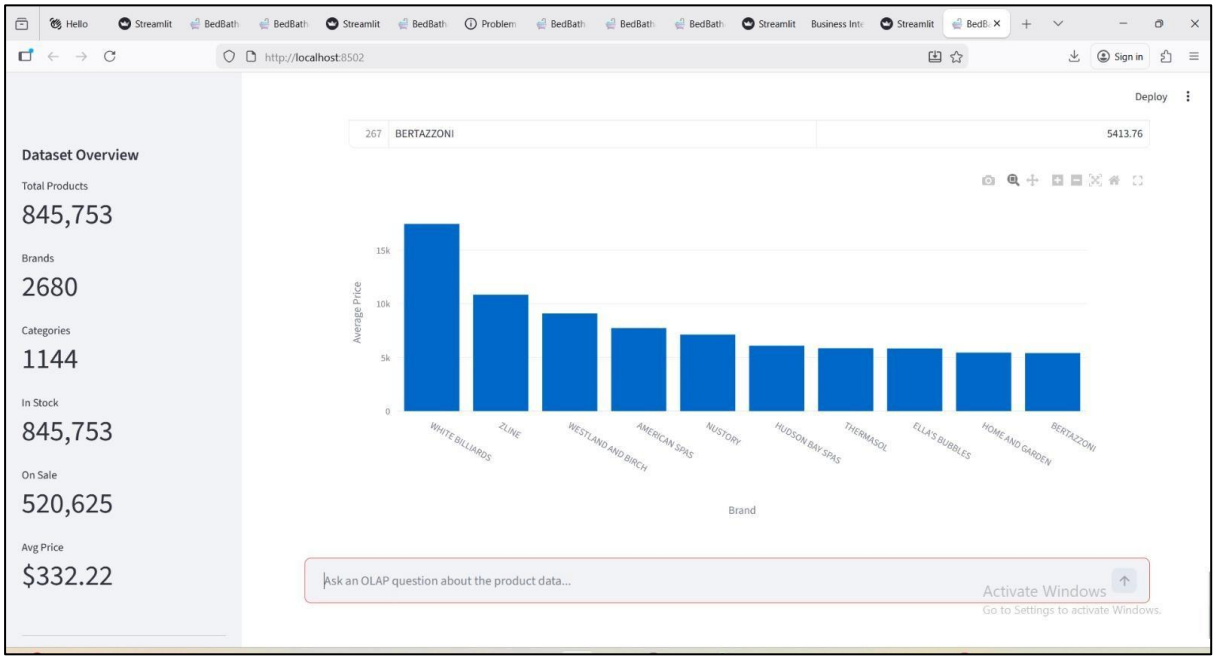
Operation 3: GROUP & SUMMARIZE

Aggregates the dataset by a dimension and calculates summary statistics. Returns a bar chart automatically.



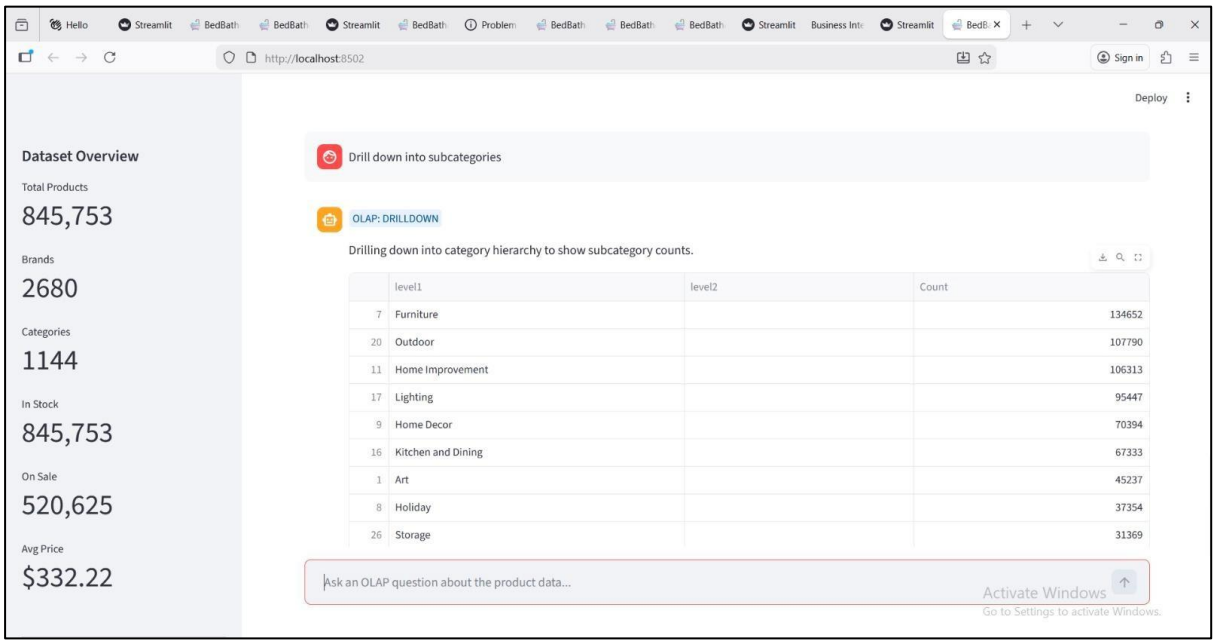
Operation 4: DRILL-DOWN

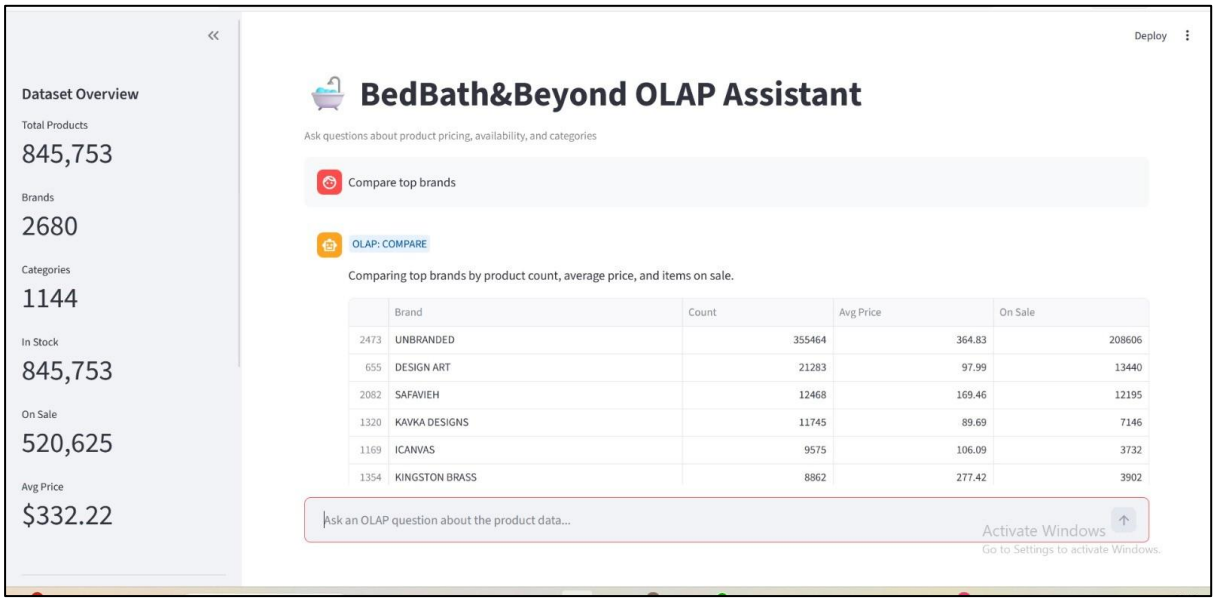
Navigates from a summary level down to a more detailed level by parsing the category breadcrumb hierarchy.

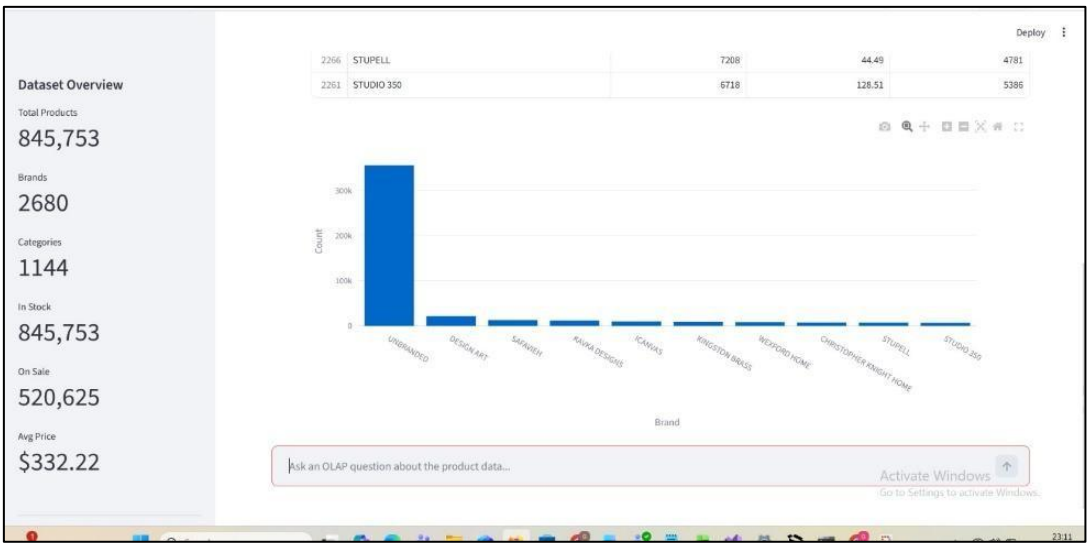


Operation 5: COMPARE

Places multiple dimension values side by side for comparison across one or more measures.



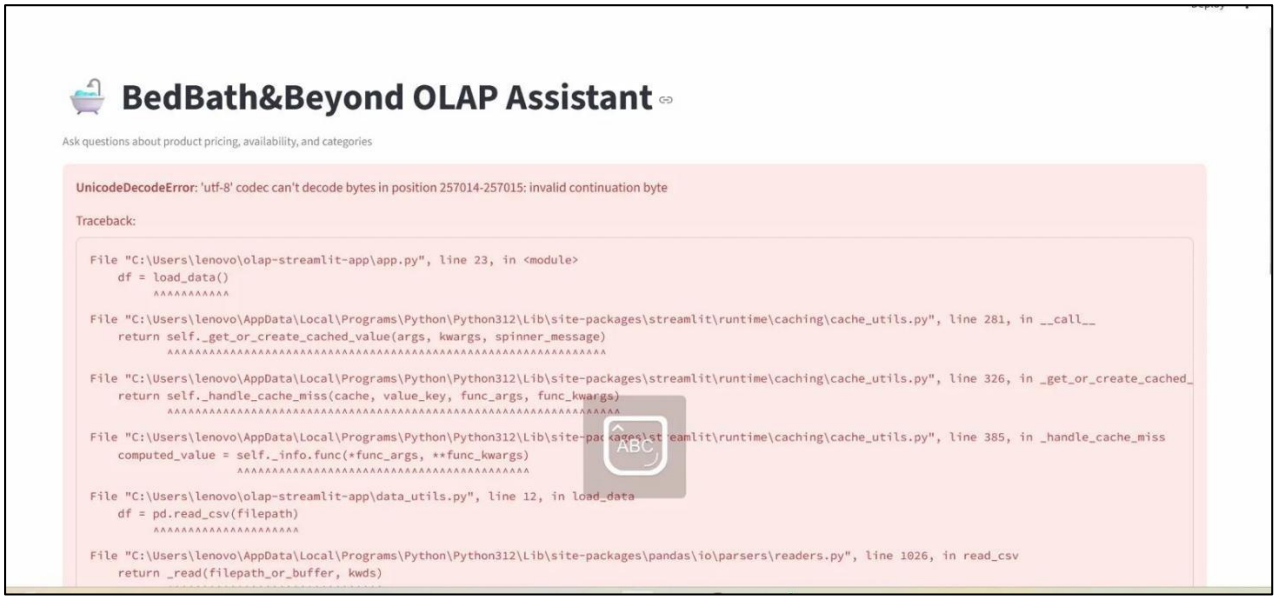




8. Technical Challenges and Solutions

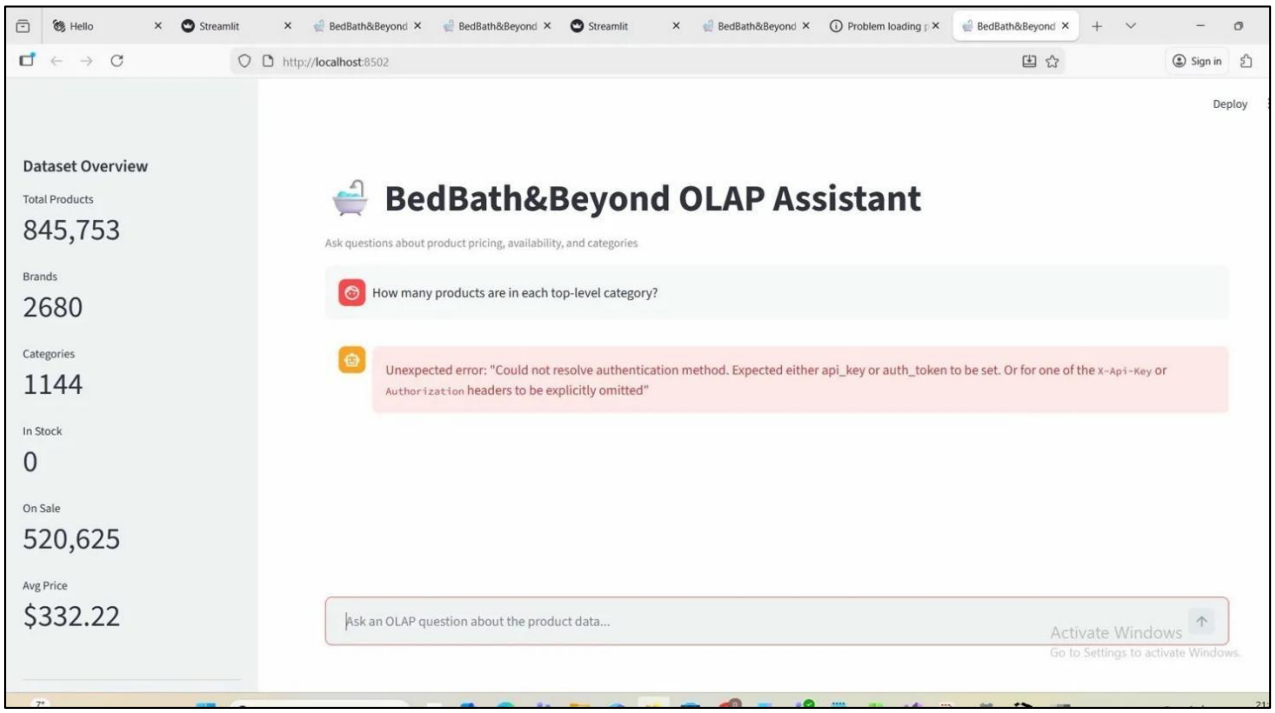
Challenge	Solution
CSV encoding error (UnicodeDecodeError)	Added encoding='latin-1' and low_memory=False parameters to pd.read_csv()
secrets.toml saved with wrong encoding (UTF-16)	Rewrote file using Python with explicit UTF-8 encoding
Column names mismatch between prompt and real data	Ran python -c to inspect actual column names, updated all references
Indentation errors when pasting code into Notepad	Rewrote data_utils.py using PowerShell WriteAllText with correct indentation
API authentication failure	Corrected secrets.toml format and verified key validity on console.anthropic.com

9. Error Handling



Data Processing Error Handling

This figure presents an encoding error occurring during dataset loading in the OLAP application. The displayed traceback supports debugging and helps resolve data quality issues before analysis.



Authentication Error Handling

This figure shows the OLAP Assistant detecting missing API authentication credentials. The system displays an error message to help identify configuration problems and ensure secure operation.

CONCLUSIONS

1. Key Findings:

Data Pipeline Success The project successfully processed 845,753 BedBath&Beyond products through an 8-step cleaning pipeline, resulting in a standardized, analysis-ready dataset. The pipeline resolved major quality issues including inconsistent labels, null values, and unstructured text fields.

Data Quality Issues Uncovered The raw data suffered from four main problem categories: missing values (null brands, absent sale prices), inaccuracies (unrealistic prices, misspellings), inconsistencies (varying availability labels, formatting differences), and incompleteness (missing descriptions and specifications).

Prompt Engineering as a Core Engineering Discipline the most significant insight from Phase 3 was that AI behavior design is iterative and engineering-like. Early failures stemmed from mismatched column names, missing JSON format enforcement, and ambiguous OLAP definitions — each requiring targeted prompt fixes.

Phase Interdependence Data quality work in Phases 1–2 directly determined the reliability of AI-generated analysis in Phase 3. Clean, standardized column names and null handling made pandas code generation by Claude significantly more accurate.

2. Improvements to Address existing limitations

AVAILABILITY Field Capitalization Bug The in-stock counter shows 0 due to a case mismatch. A simple .str.upper() normalization at load time in data_utils.py would fix this immediately.

Session Memory Each session starts fresh. Storing conversation history in Streamlit's session_state and passing it to the Claude API would enable meaningful follow-up questions.

Error Handling for Generated Code Claude occasionally produces incorrect pandas code for complex queries. Adding a retry mechanism with the error message fed back into the API call would allow self-correction without crashing the app.

3. Future enhancements

- **Near-Term**
  - Deploy to Streamlit Cloud for browser-accessible use without local Python setup.
    Add Excel/PDF export of analysis results.
  - Implement result caching to avoid re-running identical queries on the large dataset.

- **Medium-Term**
  - Add time-series tracking to support Price Monitoring use cases (requires scheduled re-scraping).
  - Introduce user-defined filters in the sidebar (brand, category, language) to pre-scope queries before sending to Claude.

- **Longer-Term**
    - Implement a Tier 3 multi-agent architecture where specialized agents handle distinct OLAP operations in parallel, improving accuracy on complex multi-step queries.
    - Expand to a multi-source pipeline by merging competitor datasets, using the WEBSITE and CLIENT constants already added in Step 6 of the pipeline.
    - Add a semantic search layer so users can find products by description similarity rather than only structured queries.

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